

Debasish Chakroborti (Joy), Ph.D.

 <https://joydeba.github.io/>  debasish.chakroborti@usask.ca  <https://github.com/joydeba>
 <https://www.linkedin.com/in/joydeba/>  <https://scholar.google.com/citations?user=VdrsyT8AAAAJ>
 <https://orcid.org/0000-0002-1597-8162>  <https://stackoverflow.com/users/1797926/joydeba>  @joydeba
 410 Stensrud Rd #118, Saskatoon  306 914 6212

Skills

Languages	English and Bengali (strong reading, writing, and speaking).
Coding	Python, Java (J2SE, J2ME), C++, C#, Ruby, JavaScript, R, PHP, MATLAB, PROLOG, OpenGL, Assembly (8085, 8086 MASM), SQL, $\text{\LaTeX}$.
Frameworks	Django, Rails, Flask, ASP.NET, AngularJS, WordPress, Joomla, Spark, Hadoop, TensorFlow/PyTorch.
Databases	MySQL, PostgreSQL, SQL Server, CouchDB, SQLite.
Web	HTML, CSS, JavaScript, Ajax, XML, Apache Web Server, Git, GitHub, SVN.
Systems	Windows, Linux (Ubuntu), macOS. DNS and mail server configuration.
Focus	Software engineering, AI/ML for code, backporting, big data, teaching.

Certifications & Professional Development

AWS	AWS Academy Educator. AWS Skill Builder digital badges / micro credentials (2024 to 2025), including AWS Cloud Practitioner Essentials, Designing Blockchain Solutions using Amazon Managed Blockchain, Amazon S3 Business Continuity and Disaster Recovery, AWS Managed Services (AMS) Advanced Disaster Recovery, Getting Started with Amazon ECS, Getting Started with DevOps on AWS, Amazon EKS Primer, Amazon EMR Getting Started, AWS Backup Primer, and AWS Billing and Cost Management.
Teach	LIFT (Learners, Instruction & Fundamentals of Teaching). Statement of Achievement, 90 hours (Saskatchewan Polytechnic, May 2026). New Instructor Orientation (NIO) . Statement of Achievement, 18 hours (14 to 16 Jan 2025). Reconciliation Education . Certificate of Completion, First Nations University of Canada (Indigenous awareness for the workplace). 4 Seasons of Reconciliation PD. Staff version (Oct 2025). Learning & Teaching Kickoff Day . Statement of Attendance (Aug 2024). WHMIS 2015 . Saskatchewan Polytechnic (issued Nov 2024, valid to Nov 2027).
CETL	1-day Training on Pedagogy (Induction for the Beginners). Centre of Excellence in Teaching & Learning (CETL) and Institutional Quality Assurance Cell (IQAC), Khulna University (HEQEP), 12 September 2014. 3-day Training on Pedagogy (Module I Teaching & Learning). Centre of Excellence in Teaching & Learning (CETL) and Institutional Quality Assurance Cell (IQAC), Khulna University (HEQEP), 11 to 13 June 2015. 2-day Training on Pedagogy (Module II Teaching & Learning). Centre of Excellence in Teaching & Learning (CETL) and Institutional Quality Assurance Cell (IQAC), Khulna University (HEQEP), 30 to 31 December 2016.
GTI	14th Foundation Training for University Teachers (FTUT) . Graduate Training Institute (GTI), Bangladesh Agricultural University, Mymensingh (UGC sponsored). Residential foundation training for public university teachers, 150 hours , 24 May to 22 June 2014 (certificate no. 22440, nominated as Lecturer, CSE, Khulna University). Modules included Teaching Methods & Techniques (50 hours), Statistical Analysis, Research Methods and Project Cycle Management, Scientific Report Writing, Administration and Office Management, and Web Page Design and Hosting.
Betar	Industrial training , Bangladesh Betar, Khulna (5 to 26 January 2012). Radio broadcasting and control systems, magnetic tape to computerized conversion, power supply and central AC management, and internal network management.
Ethics	TCPS 2 CORE 2022 . Course on Research Ethics (Panel on Research Ethics, Certificate #0000795671, Feb 2022).
Policies	Workplace policies and compliance (acknowledged onboarding). CloudLinux Acceptable Use Policy (SOP_SEC_02), Employee Privacy Policy (SOP_DP_01), and Code of Ethics . Saskatchewan Polytechnic Code of Conduct (Policy 703) / institutional policy acknowledgment.
Coursera	Coursera . Machine Learning (Stanford University, Dec 2019). Deep Learning Specialization (deeplearning.ai, Jul 2020, five courses). Statistical Inference (Johns Hopkins University, course projects Sep 2020). Data Science Specialization coursework including Statistical Inference.
Members	ACM Student Member (membership no. 2631649, renewed through 31 Jan 2025). IEEE Member (joined Oct 2018).
HackerRank	HackerRank Role Certifications. Software Engineer (Mar 2024, ID 2179B8FECE54). Software Engineer Intern (Mar 2024, ID 668513C4226F).
Security	CloudLinux workplace security training suite (Jun 2025), covering cybersecurity essentials, anti phishing, social engineering, insider threats, data protection, remote work information security, and data breach response modules.

Employment History and Projects

- 2025/26 ■ **Consultant (AI Automation Team), Cloud Linux Software, Inc., USA.** Research advisory on AI based automated backporting tooling.
- 2024/Pres. ■ **Instructor, School of Computing and Digital Innovation, Saskatchewan Polytechnic (Regina Campus).** Computer Systems Technology and Cloud Computing & Blockchain. Selected faculty participant, China Henan (ZJTIE) Transnational Education Project (2026): teaching and conducting labs in person in China, faculty meetings for collaboration, cultural exchange, and preparation for faculty travel abroad (institutional travel training and partnership coordination).
- 2024/Pres. ■ **Living Skies Postdoctoral Fellow, Department of Computer Science, University of Saskatchewan.** NSERC CREATE Software Analytics Research (SOAR) project. Automated software backporting and AI driven development tools.
- 2025 ■ **Sessional Lecturer, Department of Computer Science, University of Saskatchewan.** Taught **CMPT 145.3** Principles of Computer Science (Spring/Summer 2025, MT1 and MT2).
- 2024 ■ **Software Engineer (Intern), Push Interactions.** Enhanced <https://www.wilger.net>.
- 2023 ■ **PhD Intern (Remote), Synesis IT GmbH, Germany.** API planning and big data databases for Traffic Management Intelligent Transportation Systems.
- 2022/24 ■ **GSA Office Web Coordinator (Part-time), Graduate Students' Association, UofS.** Maintained the GSA website at <https://gsa.usask.ca> and added pages for services and events. HTML, JavaScript.
- 2022/24 ■ **Software Engineer (Part-time), Saskatchewan Education Leadership Unit.** Maintained the Placement Application at <https://placement.usask.ca> and developed a matching model for Teacher Candidates and Mentors.
- 2017/24 ■ **TA / RA, M.Sc. and Ph.D., University of Saskatchewan.** Teaching Assistant / Student Assistantship (PSAC 40004) for CMPT 214 (Fall 2023), 145 (Summer 2023), 370 (Fall 2022), 280 (2022 to 2023), 270 (2020 to 2022), 141, and Database Management Systems. Plant Phenotyping and Imaging Research Centre and Global Institute for Food Security. Developed **RISP** and **RISPs** data reusability algorithms (Hadoop/Spark). Built SE and imaging tools **Baczer**, **ReBack** (CNN), **BackTrans** (LLM), **BackSlice**, and **InsCount** (UNet).
- 2017/24 ■ **Lecturer (on study leave), CSE Discipline, Khulna University, Bangladesh.** On study leave for graduate studies at the University of Saskatchewan.
- 2013/17 ■ **Lecturer, CSE Discipline, Khulna University, Bangladesh.** Teaching and research. Implemented websites for Khulna University (<https://ku.ac.bd>), ICCIT (<https://iccit.org.bd/2013/>), and the Convocation 2016 Registration Website.
- 2014/16 ■ **Part-time Lecturer, North Western University, Khulna, Bangladesh.** Computer Science courses for the BA program.
- 2014/16 ■ **Instructor (Part-time), Higher Secondary Teachers' Training Institute (HSTTI), Khulna, Bangladesh.** ICT courses for high school teachers.
- 2013 ■ **Software Engineer, Nascenia Limited, Bangladesh** (29 May to 30 September 2013). **SnapKnot** wedding photographer directory (<https://snapknot.com/>). Ruby on Rails, ASP.NET, C#, Git.
- 2012/13 ■ **Software Engineer, Divine IT Limited, Bangladesh.** ERP systems (<https://www.divineit.net>), including Lines Pay, Sphere, Prism, and Centroid. Django, MySQL, PostgreSQL, sqlite, SVN.

Selected Projects & Tools

- SE / AI ■ **Baczer**, **ReBack** (CNN), **BackTrans** (LLM), and **BackSlice**. Analysis, recommendation, and adaptation of backporting pull requests.
- Vision ■ **InsCount** (UNet). Dense organ counting in plant images for phenotyping research.
- Data ■ **RISP / RISPs**. Intermediate data reusability in scientific workflow management (Hadoop/Spark).
- Web ■ Placement Application (<https://placement.usask.ca>). GSA website (<https://gsa.usask.ca>). Wilger (<https://www.wilger.net>). SnapKnot. Khulna University / ICCIT sites. Convocation 2016 Registration Website (Khulna University). Divine IT ERP products Lines Pay, Sphere, Prism, and Centroid.
- Undergrad ■ **Simple Arithmetic Evaluation** (Java/J2SE). **Mobile Photo Editor** (J2ME). **Geographic Information System Website** (ASP.NET). **Khulna City Corporation Holding Tax Management System** (C#/.NET). **KU Alumni Website**. **E-library Website**. **Reunion website** (PHP).
- Photo ■ **ChakroLens**. Personal photography brand and toolkit (logos, outlined watermarks, and Darktable concert RAW processing styles for clean and dramatic looks). Website and social pages forthcoming.

Education

- 2020/24 ■ **Doctor of Philosophy, Computer Science/Software Engineering**, University of Saskatchewan, Canada. Thesis titled *Automated Backporting for Efficient Version Management in Software Repositories*. Advisors Prof. Dr. Chanchal K. Roy and Prof. Dr. Kevin Schneider. January 2020 to December 2024. Course mark 89%. Awards include Carl McCrosky Innovation Scholarship for Graduate Studies (2023), GSA Mark Kroeker Exceptional Student Leadership Award (2023), Ph.D. Citizenship Award (2024), Education Graduate Award from Friends Club of Saskatoon (2024) for remarkable contribution in research and innovation, Faculty/Departmental Scholarship, SOAR, SEMLA, and CSGC Best Poster Awards, Ivan and Margaret Toutloff Award, and Graduate Travel Awards (Seattle, Melbourne, Montreal).
- 2017/19 ■ **Master of Science, Computer Science/Big Data**, University of Saskatchewan, Canada. Thesis titled *An Intermediate Data-driven Methodology for Scientific Workflow Management System to Support Reusability*. Advisors Prof. Dr. Chanchal K. Roy and Prof. Dr. Kevin Schneider. September 2017 to November 2019 (thesis defended 12 November 2019, requirements completed 23 November 2019). Course mark 89%. Awards Saskatchewan Innovation and Opportunity Scholarship and University of Saskatchewan Faculty Stipend.
- 2008/12 ■ **Bachelor of Science, Computer Science and Engineering**, Khulna University, Bangladesh. Thesis titled *Web Service Performance Enhancement for Portable Devices Modifying SOAP Security Principle*. Supervisor Dr. Masud Rahman. January 2008 to May 2012. CGPA 3.73/4.00 (1st class 3rd). **Class Representative** (Term 3.1), CSE Discipline. Awards Innovation Award, Khulna University, Dean's Merit Scholarship / Merit List Scholarships (2008 to 2012), 3rd place at Digital Innovation Fair, Khulna (2010) for Geographic Information System Website and Khulna City Corporation Holding Tax Management System, and contestant of Citi Financial IT Case Competition (organized by D.Net, 2010). Coursework projects include Simple Arithmetic Evaluation (Java), Mobile Photo Editor (J2ME), Geographic Information System Website, Khulna City Corporation Holding Tax Management System, KU Alumni Website, E-library Website, and Reunion website (PHP).
- 2006 ■ **Higher Secondary Certificate (H.S.C.), Science**, Police Line School and College, Rangpur, Bangladesh (Rajshahi Board).
- 2004 ■ **Secondary School Certificate (S.S.C.), Science**, Gayabari School and College, Dimla, Nilphamari, Bangladesh (Rajshahi Board). Recognized among the **top 10 students in Dimla Upazila** for the S.S.C. result. **Winner**, Dimla Upazila general knowledge competition (Grade 9 / Class IX), as part of a **4-person team**.

Awards, Leadership & Service

- Pubs ■ **23 research items** (2015 to 2026, including under review/submitted) in venues including ICSE, ICPC, MSR, CASCON, ASE Journal, EMSE, EICS, and Big Data.
- 2023/26 ■ **Selected Faculty Participant**, China Henan (ZJTIE) Transnational Education Project, Saskatchewan Polytechnic (2026: teaching and labs in person in China, faculty collaboration meetings, cultural exchange, and travel/partnership preparation). **Ph.D. Citizenship Award 2024**. **Education Graduate Award**, Friends Club of Saskatoon (2024), for remarkable contribution in research and innovation. **Carl McCrosky Innovation Scholarship** (2023). **GSA Mark Kroeker Exceptional Student Leadership Award** (2023).
- 2017/24 ■ **Saskatchewan Innovation and Opportunity Scholarship**. Departmental Scholarship. SOAR Best Poster (2021 symposium). SEMLA / CSGC Best Poster Awards. Ivan and Margaret Toutloff Award. Graduate Travel Awards (Seattle, Melbourne, Montreal).
- 2008/12 ■ **Innovation Award**. **Dean's Merit Scholarship** / Merit List Scholarships, Khulna University (2008 to 2012). **3rd place**, Digital Innovation Fair, Khulna (2010), for Geographic Information System Website and Khulna City Corporation Holding Tax Management System. **Contestant**, Citi Financial IT Case Competition (organized by D.Net, 2010). **Honorable Mention**, AUST Inter University Programming Contest (IUPC 2009).
- 2002/04 ■ Recognized among the **top 10 students in Dimla Upazila** for the S.S.C. result (Gayabari School and College, 2004). **Winner**, Dimla Upazila general knowledge competition (Grade 9 / Class IX), as part of a **4-person team**.

Awards, Leadership & Service (continued)

Service ■ **President**, Computer Science Council / CSGC (2022 to 23). Lead Organizer, 7th ICSAC / Research Fest. Distinguished Speaker recognition, CSGC Research Fest. **Judge**, Research Fest 2025 (Computer Science Graduate Council, Department of Computer Science, USask). **Judge**, Images of Research contest, USask (2024). **Judge**, GSA Graduate Research Elevator Pitch Competition (2024). **Judge**, GSA Graduate Research Conference (2023). **Runners-up**, CSGC Game Night 29 (2024). **Champion**, Card 29, Esho Kheli (BSAUS, 2026). **Champion**, Call Bridge, Esho Kheli (BSAUS, 2026). **Runners-up**, Call Bridge, Esho Kheli (2024). VP Social / VP Admin, CSGC UofS. **President**, Huskies Cricket Club (GSA ratified social group, 2023 to 24). **Founder and Captain**, Borno Bangla cricket team. **GWC Runner up Trophy** (first season, 2025). **Best Bowler**, Short Pitch Cricket Tournament (2025). Team member, **champion** team, BCAS T10 Cricket Tournament (2023). Team member, **runners-up** Bangladeshi team, Dakota Dunes Casino Mini Cricket World Cup (2023). Team member, **runners-up** team, BCL Cricket Tournament (2022). Team member, **runners-up** team, cricket tournament organized by Tigers and BSAUS (2018). Team member, **runners-up** team, BCAS Independence Day Football Tournament (2018). **Badminton Doubles Champion**, Khulna University Teachers' Club (2015). VP Social, Bangladeshi Community Association of Saskatoon. VP Events, Bangladeshi Association USask. GSA rep on Associate Dean Students Search Committee and Arts & Science Planning Advisory Committee. **Volunteer Lead**, Digitized Technology (University of Saskatchewan outreach event for high school students). Volunteer Lead (SSPP). Volunteer (Global Village). **BCAS Volunteer**, Saskatoon Folkfest (2021 to 25). **Photographer**, Rockfest 2025 (Echoes of Bengal). **Photographer**, 1st Annual SSPP Banquet (2019). **Photographer**, 6th Annual SSPP Banquet (2026). **Photographer volunteer**, various SSPP Durga Puja Celebrations. **Class Representative** (Term 3.1), CSE Discipline, Khulna University (B.Sc.). **Volunteer**, National Seminar on Computer and Information Technology, Khulna University (2010). **Volunteer**, Inter-discipline Cyber Gaming Contest, Khulna University (2009). **Organizer/participant**, cultural programs, Khulna University (2008 to 2012).

References

■ Available upon request.